Quiz 4

Friday, September 25 — angle, nearest neighbors, and clustering

Last updated: September 20, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Related lectures: Angle, Neighbors, Clustering (Week 4). Same day: Basis (not on this quiz).

Related lab: Lab 2 (Monday, September 28; builds on Lab 1).

Textbooks: Boyd & Vandenberghe (VMLS) · Lachniet Octave PDF · also on Resources.

Schedule

Core

- B 3.4–style angle / cosine similarity (from Lecture 08)
- B 3.24; nearest-neighbor choice of distance vs angle (Lecture 09)
- B 4.1, 4.2, 4.3
- C4.1: Compute one assignment or update step of k-means on a tiny dataset.
- C4.2: Explain whether a nearest-neighbor or cluster result changes under feature rescaling (Labs 1–2 / Lectures 09–10).

Stretch

B 4.4; a second k-means iteration or a brief interpretation of a centroid.

What a quiz item might change

Three short feature vectors, which similarity measure is used, one feature's units, or the initial cluster representatives.

Note

Dependence, span, and basis (Lectures 12–14) appear on Midterm 1 (Fri Oct 2; day page) and again on Quiz 5 (Fri Oct 9) — not required for this quiz.